MILITARY SPECIFICATION

BONDING OF PRECISION MIRRORS
(Glass-to-Metal)

1. SCOPE

1.1 This specification covers the bonding (glass-to-metal) of high resolution optical mirrors (front surface) to their metal mounts for use in precision military fire control instruments.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-504	Plastic Sheet and Film, Cellulose Acetate

Military

MIL-S-11031	Sealing Compound, Adhesive: Curing (Poly-Sulfide Base)
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of

STANDARDS

Military

MIL-STD-810	Environmental Test Methods

(Copies of specifications, standards and drawings required by suppliers in connection with procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Assembly Components.

3.1.1 Optical.- The mirror shall comply with the requirements for resolution, astigmatism, parallelism, etc. specified on the applicable drawing.

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3.1.2 Metal.- The mirror mount shall comply with the requirements specified on the applicable drawing.

3.1.3 Adhesive.- Bonding materials shall be in accordance with MIL-S-11031.

3.1.4 Spacers.- Spacers used to control bonding thickness shall be 1/4" diameter by 0.025" ± 0.002" thick cellulose acetate in accordance with L-P-504.

3.2 Bonding Process.- Unless otherwise specified by the procuring agency, assemblies shall be bonded in accordance with the procedure specified herein. Deviation from this procedure is prohibited unless approval is obtained from the contracting officer prior to use in production.

3.2.1 Cleaning.

3.2.1.1 Mirror.- Polishing compounds, pencil marks and other contaminants shall be removed from the surface of the glass to be bonded by scouring with a past of pumice or calcium carbonate and water applied with cotton swab. Residue shall be removed by rinsing with distilled water and wiping with clean, dry cheesecloth followed by an ethyl alcohol rinse and wiping.

3.2.1.2 Metal Component.- The bonding surface shall be of a virgin metal free from corrosion products, burrs, passivation or oxide treatments and any other surface contaminants; such imperfections shall be removed by carefully rubbing with 1-G emery polishing paper while at the same time not destroying the physical dimensions and flatness of the bonding surface. The surface shall be wiped with ethyl alcohol saturated cheesecloth and allowed to dry.

CAUTION: Clean surfaces are essential prerequisites for obtaining a satisfactory bond.

3.2.2 Bond Configuration.- The bond shall take the form of three distinct circlets formed at the corners of an imaginary triangle located as shown on the applicable assembly drawing.

3.2.3 Bonding Fixture.- The bonding fixture shall be designed to provide mechanical positioning of components to be bonded to an accuracy compatible with the applicable drawing and to maintain a uniform pressure by spring loading. Sufficient support shall be provided to protect components being bonded from unnecessary strain. It shall accommodate easy removal of the bonded assembly without straining or bending. Since the fixture, optical and metal components and the adhesives are influenced by heat during the curing cycle, care shall be exercised to fabricate fixtures from materials which will maintain alignment and pressures.

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3.2.4 Bonding.- Bonding shall conform to the procedures outlined below. These techniques involve the basic operations of bonding with a polysulfide adhesive/sealant whose mix ration is modified for this specific application.

3.2.4.1 Adhesive Mix Ratio.- Mix the adhesive of MIL-S-11031 in the ratio of two parts base to one part accelerator. Stir thoroughly so as to insure complete dispersion of the accelerator in the base. Do not mix more material than is necessary to bond 2 or 3 assemblies at one time as the pot life of the adhesive at this mix ratio is short.

3.2.4.2 Adhesive Application.- Place the mirror in the fixture with the surface to be bonded facing up. "Puddle" the adhesive on the three distinct areas of the glass surface at the locations specified on the applicable drawing. A sufficient, but not excessive, quantity of adhesive should be deposited such that when the metal support bracket is placed in position, three separate circlets approximately 1" to 1-1/8" in diameter by 0.025" thick will be formed.

CAUTION: The circlets shall not touch one another after cure.

3.2.4.3 Spacing.- Within each puddle of adhesive centrally place the 0.025" thick spacer and press it down through the adhesive until it bottoms on the glass surface.

3.2.4.4 Metal Component Positioning and Application of Pressure.- Carefully locate the metal component on the adhesive puddles in the proper position as shown on the applicable drawing. Slowly and evenly apply pressure to the back of the component so that the adhesive flows out to a maximum of 1-1/8" diameter circle and the metal component bottoms on the spacers to yield a bond thickness of 0.025". Maintain a 5 psi pressure on the assembly during the oven cure.

3.2.4.5 Cure.- Place the fixture holding the prepared assembly in an unheated circulating air-type oven, and raise the temperature to a curing level of 160° ± 5°F. Maintain this temperature for a minimum of 4 hours. Remove fixture from oven and allow the bonded assembly to cool to room temperature. Assure that both the mirror and metal component are securely fastened and undisturbed during cure so that the assembly will meet the requirements specified on the applicable drawing.

NOTE: In some cases it is possible to remove the bonded assembly from the fixture immediately after the oven cure without affecting any optical or mechanical properties.

3.3 Performance Requirements.- The bonded assembly shall meet the alignment and optical performance requirements as specified on the applicable drawing after the following environmental conditions:

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3.3.1 Elevated Temperature Thermal Stability.- After stabilization at room temperature, exposure at 160° ± 2°F for 4 hours.

3.3.2 Low Temperature Thermal Stability.- After stabilization at room temperature, exposure at -80° ± 2°F for 4 hours.

3.3.3 Mechanical Shock.- Subsequent to 3.3.1 and 3.3.2, 200 g applied parallel to the bonded surface, repeated 3 times. The shock pulse shall have a wave form approximately a half-sine wave with a minimum time duration of 2 ± 0.5 milliseconds.

3.4 Workmanship.- Workmanship shall be of the highest quality consistent with the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Performance Requirements.- Each bonded assembly shall meet the requirements specified in 3.3 utilizing the test methods specified herein. Bonded assemblies shall be prepared in accordance with Section 3, and deviations therefrom shall not be permitted without prior approval of the contracting officer.

4.2.1 Elevated Temperature.- Thermal Stability - The bonded assembly shall be subjected to the requirement specified in 3.3.1. The testing equipment utilized for this test shall be in accordance with the "Test Facilities" of MIL-F-13926. The test chamber shall be at room temperature (+60° to +90°F) when the assembly is inserted. The temperature of the test chamber shall be raised gradually to +160°F (see Note). The assembly shall remain at this temperature for a minimum of 4 hours. At the completion of this 4 hour period the temperature of the test chamber shall be gradually reduced to room temperature. The bonded assembly shall be removed from the test chamber and subjected to the test specified in 4.2.2.

NOTE: The rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling operation of the tests in 4.2.1 and 4.2.2.

4.2.2 Low Temperature Thermal Stability.- The bonced assembly shall be subjected to the requirements specified in 3.3.2. The testing equipment utilized in this test shall conform to the requirements of 4.2.1. The test chamber shall be at room temperature when the assembly is inserted. Regulate the temperature of the test chamber to 0°F. The assembly shall be permitted to stabilize at this temperature. The temperature of the chamber

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shall then be reduced gradually to -80°F. The assembly shall remain at this temperature for a minimum of 4 hours. At the completion of this 4 hour period the temperature of the test chamber shall be gradually raised to room temperature. When the bonded assembly has been stabilized at room temperature it shall be removed from the test chamber and subjected to the test specified in 4.2.3.

4.2.3 Mechanical Shock.- The bonded assembly shall be shock tested in accordance with Procedure IV, Method 516.1 of MIL-STD-810, except that the direction and number of shocks shall be as specified in 3.3.3. The shock pulse shape, peak valve and time duration shall be in accordance with 3.3.3. The adapter utilized for mounting the bonded assembly on the shock machine shall securely hold the pivot axis in a vertical plane within 2 degrees. Upon completion of the specified shock impulses the glass and metal components of the assembly shall be examined for evidence of physical damage. Any evidence of damage shall be cause for rejection of the bonded assembly. Subsequent to the shock test the bonded assembly shall meet the performance requirements of 3.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking are not applicable to the specification.

6. NOTES

6.1 Intended Use.- Bonded mirror assemblies covered by this specification are primarily intended for those applications where minimal distortion of the mirror by the adhesive bond is required. As such, it is normally used for assemblies requiring an optical resolution better than 2 seconds of arc. This bonding method may find application for elliptical reflectors and beam splitters but should be thoroughly tested prior to use.

6.2 Ordering Data.- Procurement documents should specify the title, number, and date of this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 12GP-A032Y

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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